MILITARY SPECIFICATION

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-20G, dated 5 November 1976, and is approved for use by all Departments and Agencies of the Department of Defense.

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TABLE II, symbols F and G, delete and substitute:

Symbol	Capacitance tolerance
	(±)
F - - - - - - - - - - 1/	1 percent
G - - - - - - - - - - 1/	2 percent

After table II, add footnote 1/ as follows:

"1/ For nominal capacitance values of 10 pF or less, the capacitance tolerances shall be ±1.0 pF (F) and ±2.0 pF (G)."

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3.2, line 3: At end of paragraph, add "Unless procured from the manufacturer or his authorized distributor listed or approved for listing on the qualified products list, ER parts furnished under this specification shall not be considered as having met the requirements of this specification."

3.6, delete and substitute:

"3.6 Thermal shock and voltage conditioning (ER parts only). When tested as specified in 4.7.2, capacitors shall withstand the extremes of high and low temperatures without visible damage and meet the following requirements:

Dielectric withstanding voltage (at 25°C) - - - - - - - - - - - -	As specified in 3.11.
Insulation resistance (At 25°C) - - - - - - - - - - - - - - - - -	Shall be not less than the value shown on figure 2.
(At 125°C)- - - - - - - - - - - - - - - - - -	Shall be not less than the value shown on figure 2.
Capacitance (at 25°C) - - - - - - - - - - - -	Shall be within the tolerance specified (see 3.1).
Dissipation factor (at 25°C)- - - - - - - -	Shall not exceed the initial requirement (see 3.9)."

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* 3.23.1 and 3.23.1.1, delete and substitute:

"3.23.1 Method I. Marking of capacitors shall conform to method I of MIL-STD-1285. Unless otherwise specified (see 3.1), capacitor marking shall include the type designation, 'JAN' brand (ER only), trademark or manufacturer's name, source code, voltage, date code, and lot symbol in the order shown (see example).

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Example of ER:

CCR78CG - Type designation (see 1.2.1).
821KM
JAN TM - 'JAN' brand and trademark or manufacturer's name.
12345 - Source code (Commercial and Government Entity, (CAGE)).
200V 7803A - Voltage, date code, and lot symbol.

Example of Non-ER:

CC78CG - Type designation (see 1.2.1).
821K
TM - Trademark or manufacturer's name.
12345 - Source code (Commercial and Government Entity, (CAGE)).
200V 7803A - Voltage, date code, and lot symbol.

"3.23.1.1 Type designation marking. Unless otherwise specified (see 3.1), the type designation shall be divided between the characteristic and capacitance symbols. An ER part may be marked and furnished as a non-ER part, if produced on the same assembly line, and provided it is subjected to and meets all the inspection requirements of the ER part."

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TABLE IV, colors Black and White, delete and substitute:

Color	Characteristic 1/	Nominal capacitance (pF)		Capacitance tolerance	
		First and second significant figures	Multiplier 2/	For nominal capacitances greater than 10 pF (±)	For nominal capacitances of 10 pF or smaller (±)
" Black- - -	C-	0	1	- - -	2.0 pF (G)
White- - -	- - -	9	0.1	10% (K)	1.0 pF (F) "

* After 3.24, add 3.25 and 3.26 as follows:

"3.25 Marking legibility (laser marking only). When tested as specified in 4.7.19, the marking shall remain legible.

"3.26 Supplying to higher FR levels. A manufacturer may supply to all higher failure rate levels than that to which he is qualified. Parts qualified and marked to lower failure rate levels are substitutable for higher failure level parts, and shall not be remarked unless specified in the contract or purchase order."

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4.1.1, delete in its entirety and substitute:

"4.1.1 Reliability assurance program (applicable to ER parts only). A reliability assurance program shall be established and maintained in accordance with MIL-STD-790; 'Program implementation' exceptions are as follows:

a. Under 'Description of production processes and controls', the procedure for identification of each production lot shall include only 'the manufacturer shall as a minimum be able to identify the time period during which the final production operation was performed on each item of product prior to final test. The date or lot code marked on each part shall be identified to a production lot.'

b. 'Traceability' of materials shall not apply.

Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification."

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